Grade 10 Chemistry

Solved Exercises - English Medium - Balochistan Textbook Board (BTBB)

Complete book: all 8 chapters (9-16), every chapter-end Exercise (fill in the blanks, multiple choice, brief/short, reasoning and descriptive questions).

Chapter 9: Chemical Equilibrium

Chapter 10: Acids, Bases and Salts

Chapter 11: Organic Chemistry

Chapter 12: Hydrocarbons

Chapter 13: Biochemistry

Chapter 14: Environmental Chemistry-I: The Atmosphere

Chapter 15: Environmental Chemistry-II: Water

Chapter 16: Chemical Industries

Source: BTBB Grade 10 Chemistry textbook, 190 pages. Medium confirmed from the book itself: cover and Chapter 9 lesson (book p.7 = PDF p.8) are in English. Exercises were read from rendered pages (book pp.18-20, 42-44, 73-75, 104-106, 126-128, 141-142, 156-157, 170-172). Questions are reproduced from each chapter Exercise; answers are written in medium length. Book quirks noted honestly (answers follow the book's intended option; see the manifest for detail): Chapter 10 MCQ vii: the book prints the exponents without minus signs; read as [H+] = 10^-5 giving [OH-] = 10^-9, so option (c) is ticked for its magnitude. Chapter 13 counts 10 essential amino acids (fill-in v and MCQ vi agree); the standard count is 9, but the book is followed. Chapter 14 MCQ x: the book's intended answer is 0% O2 in the thermosphere (oxygen there is atomic, not O2 molecules). Chapter 15 MCQ iv: the book's intended answer is that safe drinking of hard water depends upon the type of impurity. Chapter 15 fill-in v: 68.7% is the book's own figure for glaciers and polar ice caps combined (book p.144); it is used for the Antarctic ice sheet item. Chapter 15 fill-in vii: the exact two-word completion is not stated verbatim in the lesson, so it is filled with the chapter's own named dissolved pollutants (heavy metals and organic toxins, section 15.7). Chapter 15 fill-in viii: gastro is attributed to contaminated water, following section 15.7.2, which lists GIT diseases among those carried by household debris in water. Chapter 16 fill-in v (petroleum found in more depth than water) and viii (ammonium carbamate as the hydrated form of urea, which dehydrates to urea) are the best readings of the book's phrasing. Chapter 16 exercise pages are book pp.170-172; the contents page number 173 is where the Glossary begins.

Chapter 9: Chemical Equilibrium

Exercise (Chapter 9)

A. Fill in the blanks. (Filling word shown in red.)

(i) There are **two** types of equilibrium.

(ii) Vapour pressure is an important example of **physical** equilibrium.

(iii) The equilibrium constant shows **ratio** of products and reactants in a dynamic chemical process.

(iv) All chemical reactions show **dynamic** type of equilibrium.

(v) The Law of mass action was put forward by **Guldberg and Waage** in **1864**.

(vi) All natural cycles show **dynamic** type of equilibrium.

(vii) NH3 gas is used for manufacture of **urea** fertilizer.

(viii) In gaseous systems the **partial pressure** is taken in consideration instead of concentrations.

(ix) The value of Kc is **independent** of initial concentrations of reactants.

(x) In the gaseous systems **Kc** is replaced by Kp.

B. Choose the correct answer. (Correct option marked with a green tick.)

(i) The types of equilibrium found in nature could be of

(a) 2 ✓

(b) 3

(c) 4

(d) 5

(ii) The type of equilibrium found in fulcrum and fauna is

(a) Static

(b) Dynamic ✓

(c) Depends on conditions

(d) Shows no equilibrium

(iii) Most chemical reaction show

(a) Reversibility ✓

(b) Static equilibrium

(c) Heat energy

(d) Pressure

(iv) The value of Kc is dependent upon the

(a) Only on initial concentration of reactants

(b) Only on initial concentration of products

(c) Final concentration of reactants

(d) Does not depends upon initial concentration of reactants and products ✓

(v) The value of Kc can be used

(a) To increase the products of a chemical reaction.

(b) To predict the reaction pathway ✓

(c) To reduce the heat supply of system

(d) To calculate the pressure of system

(vi) The Kc is obtained by the data of

(a) initial concentration of reactants

(b) Initial concentration of products

(c) Concentration of both the reactants and the products at equilibrium state. ✓

(d) Final concentration of products.

(vii) The Value of Kc depends upon:

(a) Temperature of system ✓

(b) Pressure of system

(c) Initial concentration of reactants

(d) Concentration of reactants and products at equilibrium

(viii) Pressure of system effects the equilibrium of

(a) Liquids

(b) Solids

(c) Gases ✓

(d) All of these

(ix) The Carbon dioxide cycle involves

(a) Static equilibrium

(b) Dynamic equilibrium ✓

(c) Does not related with equilibrium

(d) Related with atmospheric pressure

(x) If for the general reaction A + B ⇌ C + D increase in concentration of D will result in the:

(a) Decrease in concentration of A

(b) Decrease in concentration of B

(c) Decrease in concentration of C

(d) Increase in the concentration of A and B. ✓

C. Brief / short questions.

(i) What types of equilibrium are there in nature?

There are two types of equilibrium in nature: static equilibrium, in which there is no visible change and nothing moves, and dynamic equilibrium, in which the forward and reverse processes continue at equal rates so that the overall amounts stay constant. Most chemical and natural cycles show dynamic equilibrium.

(ii) Write down the equilibrium constant expression for the reaction. NH3(g) +

HCl(g) ⇌ NH4Cl(s)

For NH3(g) + HCl(g) ⇌ NH4Cl(s), the equilibrium constant is Kc = 1 / ([NH3][HCl]). The solid product NH4Cl is left out of the expression because the concentration of a pure solid stays constant and is absorbed into the value of Kc.

(iii) What are the units of Kc?

Kc has no single fixed unit. Its units depend on the reaction: they are (mol/dm3) raised to the power (moles of gaseous or dissolved products minus moles of gaseous or dissolved reactants). For a reaction with equal moles on both sides Kc is unitless, and for other reactions it carries units such as mol/dm3 or dm3/mol.

(iv) Explain the term Dynamic equilibrium?

Dynamic equilibrium is the state of a reversible process in which the forward and reverse reactions continue to take place, but at exactly equal rates. Because the two rates are equal, the concentrations of reactants and products stop changing, even though the reaction itself has not stopped.

(v) Define the term 'rate of a chemical reaction'.

The rate of a chemical reaction is the change in concentration of a reactant or a product per unit time. It tells how fast a reaction proceeds, for example how quickly a reactant is used up or a product is formed, and it is usually expressed in mol/dm3/s.

D. Reasoning and descriptive questions.

(i) What do you understand by the equilibrium? Describe the types of equilibrium with examples.

Equilibrium is the state of a system in which the opposing processes balance each other so that no net change is observed. In static equilibrium nothing appears to move or change, for example a book resting on a table or a balanced see-saw. In dynamic equilibrium the forward and reverse processes both continue at equal rates, for example water evaporating and condensing at the same rate in a closed bottle, or the carbon dioxide cycle in nature.

(ii) Explain the reversible reaction with examples.

A reversible reaction is one that can proceed in both directions: reactants form products, and products can react back to give reactants again, shown by the sign ⇌. Examples are the formation of ammonia, N2 + 3H2 ⇌ 2NH3, the dissociation of ammonium chloride on heating, NH4Cl ⇌ NH3 + HCl, and the reaction of hydrogen with iodine, H2 + I2 ⇌ 2HI.

(iii) Describe the law of mass action and derive an expression for the equilibrium constant.

The law of mass action, put forward by Guldberg and Waage in 1864, states that the rate of a chemical reaction is proportional to the product of the active masses (molar concentrations) of the reacting substances. For aA + bB ⇌ cC + dD, the forward rate is kf[A]^a[B]^b and the reverse rate is kr[C]^c[D]^d. At equilibrium the two rates are equal, so kf/kr = [C]^c[D]^d / ([A]^a[B]^b). This constant ratio kf/kr is the equilibrium constant

Kc, so Kc = [C]^c[D]^d / ([A]^a[B]^b).

(iv) What do the Kc means? Explain its importance and applications.

Kc is the equilibrium constant of a reaction: the ratio of the equilibrium concentrations of products to reactants, each raised to its stoichiometric coefficient. Its importance is that it tells the extent of a reaction: a large Kc means the equilibrium lies far to the products side, and a small Kc means it lies to the reactants side. It is used to predict the direction of a reaction, to calculate equilibrium concentrations, and to judge how changes in concentration, pressure or temperature will shift the equilibrium.

(v) In the reaction between: $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ How you can increase the production of SO_3?

The production of SO_3 can be increased by shifting this equilibrium to the right. This is done by increasing the concentration of SO_2 or O_2, by increasing the pressure, since the forward reaction reduces the number of gas molecules from 3 to 2, and by lowering the temperature, because the forward reaction is exothermic. Removing SO_3 as it forms also drives the reaction forward.

Chapter 10: Acids, Bases and Salts

Exercise (Chapter 10)

A. Fill in the blanks. (Filling word shown in red.)

(i) Acids have **sour** taste.

(ii) Bases turn red litmus **blue**.

(iii) Stomach has **hydrochloric** acid which helps in digesting the food.

(iv) Salts are produced by the reaction of **an acid** with a base.

(v) The basic character of ammonia is well explained by **Bronsted-Lowry** concept.

(vi) Arhenius described acids as **proton** donor.

(vii) According to Lewis concept the **bases** tend to donate electron pair.

(viii) The aluminium chloride falls under the type of **acidic** salts.

(ix) The citrus fruits contain **citric** acid.

(x) The -log of [H+] is called the **pH**.

B. Choose the correct answer. (Correct option marked with a green tick.)

(i) A solution having pH of 7.5 mean that it is

(a) Acidic

(b) Basic ✓

(c) Neutral

(d) Amphoteric

(ii) A liquid has a pH of 7, what does this tell about the liquid?

(a) It is acid solution

(b) It is neutral ✓

(c) It is basic solution

(d) It is a solution of a NaOH

(iii) Which of the statements below is correct?

(a) All bases dissolve in water

(b) All acids are soluble in water

(c) Sodium chloride is a basic salt

(d) Alkalis are bases. ✓

(iv) Name the salt that forms when HCl is reacted with Ca(OH)2

(a) Calcium carbonate

(b) Calcium hydrate

(c) Calcium chloride ✓

(d) Calcium oxide

(v) Which pair of substances will react together to form CuSO4.

(a) Copper and Sulphuric acid.

(b) Copper oxide and Sulphuric acid ✓

(c) Copper oxide and hydrochloric acid

(d) Copper and Sodium sulphate.

(vi) The colour of potassium dichromate salt is:

(a) Yellow

(b) Green

(c) Orange ✓

(d) Blue

(vii) For a system where [H+] = 10^5, the [OH-] = ?

(a) 10^5

(b) 10^6

(c) 10^9 ✓

(d) 10^10

(viii) Phenolphthalein indicator shows purple colour in basic media, while its colour in acidic media is:

(a) Red

(b) Blue

(c) Yellow

(d) Colourless ✓

(ix) Salts are produced when a base reacts with

(a) An Acid ✓

(b) Another base

(c) Another salt

(d) Oxides

(x) CaCO3 is a salt which is:

(a) Soluble in water

(b) Insoluble in water ✓

(c) Sparingly soluble in water

(d) Soluble in benzene

C. Brief / short questions.

(i) Write down the general properties of acids.

Acids have a sour taste, turn blue litmus red, and react with metals to release hydrogen gas. They react with bases to form salts and water, and with carbonates to release carbon dioxide gas. In water they give hydrogen ions, H+, and their solutions conduct electricity.

(ii) What are the general characters of bases?

Bases have a bitter taste and a soapy, slippery feel, and they turn red litmus blue. They react with acids to form salts and water in a neutralization reaction. Soluble bases are called alkalis, and in water they give hydroxide ions, OH-.

(iii) Describe the general composition of a salt.

A salt is an ionic compound made of two parts: a positive part (cation) that comes from a base, and a negative part (anion) that comes from an acid. For example, in sodium chloride, NaCl, the sodium ion Na+ comes from the base NaOH and the chloride ion Cl- comes from the acid HCl.

(iv) What is a precipitate?

A precipitate is an insoluble solid that forms and separates out of a solution during a chemical reaction. For example, when solutions of silver nitrate and sodium chloride are mixed, white silver chloride precipitates out of the solution.

(v) What is meant by neutralization reaction?

A neutralization reaction is the reaction between an acid and a base to form a salt and water, so that the acidic and basic properties cancel each other. For example, HCl + NaOH → NaCl + H2O. The resulting solution is neutral if the acid and base are used in the right amounts.

(vi) What are the polyprotic acids?

Polyprotic acids are acids that can donate more than one proton (H+ ion) per molecule. For example, sulphuric acid, H2SO4, can donate two protons, so it is diprotic, and phosphoric acid, H3PO4, can donate three protons, so it is triprotic.

(vii) How you will explain the basic character of NH3?

Ammonia, NH3, is basic because its nitrogen atom has a lone pair of electrons that can accept a proton (H+) from an acid, as explained by the Bronsted-Lowry concept, or donate the electron pair, as explained by the Lewis concept. For example, NH3 + HCl → NH4Cl, where ammonia accepts a proton to become the ammonium ion, NH4+.

(viii) What is an indicator?

An indicator is a substance that changes colour according to whether a solution is acidic or basic, and so shows the nature of the solution. Litmus, phenolphthalein and methyl orange are common indicators: litmus is red in acid and blue in base, while phenolphthalein is colourless in acid and pink in base.

(ix) Write some uses of NaCl.

Sodium chloride, common salt, is used as a seasoning and preservative for food, and for making chemicals such as sodium hydroxide, chlorine and washing soda. It is used to melt ice on roads in winter, in tanning leather, and in the body it maintains the balance of body fluids.

(x) Name some salts that are important to medicinal use?

Salts of medicinal use include magnesium sulphate (Epsom salt), used as a laxative and in baths, and sodium bicarbonate (baking soda), used as an antacid for acidity. Potassium permanganate is used as an antiseptic, calcium carbonate is used as an antacid and calcium supplement, and plaster of Paris, calcium sulphate, is used for setting broken

bones.

D. Reasoning and descriptive questions.

(i) The Stomach contents of animals have very high acidic pH (about 1-2), explain why it does not damages the walls of stomach?

The stomach wall is not damaged because it is protected by a thick layer of mucus secreted by the stomach lining, which stops the acid from reaching the wall cells. The wall cells are also replaced very quickly, and the acid is needed in the stomach to kill germs in food and to activate the enzymes that digest proteins.

(ii) Why the acidic character of a solution increases with the decrease in pH?

The pH scale is a logarithmic measure of hydrogen ion concentration: pH = -log[H+]. A lower pH therefore means a higher concentration of H+ ions in the solution, and it is the H+ ions that give a solution its acidic character. So as pH falls, [H+] rises and the solution becomes more acidic.

(iii) Name 5 acids that are found in our food?

Five acids found in food are: citric acid in citrus fruits such as lemons and oranges, acetic acid in vinegar, lactic acid in yoghurt and sour milk, tartaric acid in grapes and tamarind, and malic acid in apples.

(iv) How salts affect our health?

Salts affect our health in both helpful and harmful ways. The body needs salts such as sodium chloride and potassium salts for nerve signals, muscle movement and fluid balance, and calcium salts for bones and teeth. But too much salt, especially sodium, raises blood pressure and harms the heart and kidneys, while a lack of iodine in salt causes goitre.

(v) How molarity is related with the pH of a solution?

Molarity gives the concentration of the acid or base in mol/dm3, and for a strong acid the hydrogen ion concentration [H+] is equal to that molarity, so pH = -log(molarity). For example, a 0.01 mol/dm3 solution of HCl has [H+] = 0.01, so its pH = 2. A higher molarity of acid therefore gives a lower pH, and for a base the pOH is found the same way and pH = 14 - pOH.

(vi) Write a note on strength of acids and bases?

The strength of an acid or base is how completely it ionizes in water. Strong acids, such as HCl, H2SO4 and HNO3, ionize almost completely, while weak acids, such as acetic acid and citric acid, ionize only partly. Similarly, strong bases such as NaOH and KOH ionize fully, while weak bases such as ammonia ionize only partly. Strength is different from concentration: a dilute strong acid can be less acidic than a concentrated weak acid.

(vii) Name a household consumer product that contains. a) A strong acid b) A weak acid c) A strong base d) A weak base

a) A strong acid: toilet cleaner containing hydrochloric acid. b) A weak acid: vinegar, which contains acetic acid, or lemon juice, which contains citric acid. c) A strong base: drain

cleaner or caustic soda (NaOH) solution. d) A weak base: ammonia-based window cleaner, or baking soda (sodium bicarbonate) solution.

(viii) Describe the theory of Arhenius to explain the acids and bases.

Arrhenius theory says that an acid is a substance that gives hydrogen ions, H+, in water, for example HCl → H+ + Cl-, and a base is a substance that gives hydroxide ions, OH-, in water, for example NaOH → Na+ + OH-. An acid and a base neutralize each other because H+ and OH- combine to form water, H+ + OH- → H2O. This theory explains aqueous solutions well, but it cannot explain the basic character of ammonia, which has no OH- group, or reactions that do not take place in water.

(ix) How Bronsted-Lowry theory helps us to understand the Acids and Bases?

The Bronsted-Lowry theory defines an acid as a proton (H+) donor and a base as a proton acceptor, so it works for any solvent, not only water. It explains why ammonia is a base: NH3 accepts a proton to become NH4+. It also explains conjugate acid-base pairs: when an acid donates a proton it becomes its conjugate base, and when a base accepts a proton it becomes its conjugate acid, for example HCl/Cl- and NH4+/NH3.

(x) Comment that the Lewis theory can be best in order to understand the nature of acids and bases.

The Lewis theory is the broadest of the three theories: an acid is an electron-pair acceptor and a base is an electron-pair donor. It covers all the cases explained by Arrhenius and Bronsted-Lowry, and also explains reactions where no proton is transferred at all, for example BF3 accepting an electron pair from NH3 to form a coordinate bond. Because it explains the widest range of acid-base behaviour, it is considered the best general theory.

(xi) What do you understand by the chemical term "Indicator"? Name some of the natural substances that contain indicators?

An indicator is a substance that changes colour with the pH of a solution and so shows whether it is acidic or basic. Natural substances containing indicators include red cabbage juice, which is red in acid and green-yellow in base, beetroot juice, turmeric, which turns red in base, and petals of many flowers such as rose and hibiscus, whose pigments change colour with pH.

(xii) Write down the colours of following indicators in both the acidic and basic medias? Phenolphthalein, Litmus, methyl orange

Phenolphthalein is colourless in acidic medium and pink (the book prints purple) in basic medium. Litmus is red in acidic medium and blue in basic medium. Methyl orange is red in acidic medium and yellow-orange in basic medium.

(xiii) Can you suggest two methods by which the sea water can be desalinated to make it drinkable?

Two methods are distillation, in which seawater is boiled and the steam is condensed to give pure water, leaving the salts behind, and reverse osmosis, in which seawater is forced

under pressure through a semi-permeable membrane that lets water molecules pass but holds back the dissolved salts.

(xiv) What is a Salt? Describe the general characters of the Salts.

A salt is an ionic compound formed when an acid reacts with a base, made of a positive ion from the base and a negative ion from the acid. Salts are usually crystalline solids with high melting points, they dissolve in water to give ions and conduct electricity in solution or when molten, and they are neutral in nature unless they hydrolyse, as acidic salts like NH4Cl or basic salts like Na2CO3 do.

(xv) How Salts are prepared and purified?

Salts are prepared by neutralization of an acid with a base, by reaction of an acid with a metal, a metal oxide or a carbonate, and by precipitation when two salt solutions react. The salt is purified by crystallization: the solution is evaporated to the crystallization point, cooled so pure crystals form, and the crystals are filtered and dried.

(xvi) What types of Salts are found in nature? Explain with examples.

The salts found in nature are normal salts, such as NaCl and KNO3, which have no replaceable hydrogen or hydroxyl group; acidic salts, such as NaHSO4 and NaHCO3, which still contain a replaceable hydrogen; basic salts, such as Cu(OH)Cl, which still contain a hydroxyl group; double salts, such as potash alum; and complex salts, such as potassium ferrocyanide. Seawater, rock salt deposits and mineral ores are natural sources of salts.

Chapter 11: Organic Chemistry

Exercise (Chapter 11)

A. Fill in the blanks. (Filling word shown in red.)

(i) Organic chemistry deals with the study of **carbon compounds**.

(ii) Wohler was the first Scientist who prepared an organic compound named **urea**.

(iii) The long chain formation ability of carbon is termed as **catenation**.

(iv) Alkanes are **saturated** hydrocarbons.

(v) A **functional group** is the group of atoms, which determines the properties of organic compounds.

(vi) General representation of the Alcohol is **R-OH**.

(vii) Ketone has the general formula **R-CO-R**.

(viii) The functional group present in Acetic acid is **carboxyl group (-COOH)**.

(ix) The % of Carbon in **anthracite** coal is 80%.

(x) The general formula of ether is **R-O-R**.

B. Choose the correct answer. (Correct option marked with a green tick.)

(i) $NaHCO_3$ is categorized as:

(a) Organic compound

(b) Inorganic compound ✓

(c) Some time organic, some time inorganic

(d) Bio-organic compound

(ii) The Hydrocarbons are soluble in the:

(a) Organic solvents ✓

(b) Water

(c) Both of above

(d) Non of above

(iii) When two molecules have same molecular formula but different structures, the phenomenon is called:

(a) Catenation

(b) Alkylation

(c) Isomerization ✓

(d) Sublimation

(iv) Organic compounds undergo chemical reactions with the

(a) Very fast rate

(b) Very slow rate ✓

(c) Neither fast nor slow

(d) They do not react

(v) The important source of organic compounds is

(a) Coal

(b) Petroleum

(c) Natural gas

(d) All of these ✓

(vi) Major compound present in the natural gas is

(a) Methane ✓

(b) Ethane

(c) Propane

(d) Butane

(vii) When one 'H' atom is removed from an alkane, it forms:

(a) Saturated hydrocarbon

(b) Unsaturated hydrocarbon

(c) Alkyl radical ✓

(d) Alkene

(viii) The fraction of petroleum that contains Carbon atoms from 4-8 is called:

(a) Gasoline ✓

(b) Kerosene

(c) Diesel

(d) Wax

(ix) The fragrance of fruits is due to the presence of functional group called:

(a) Amines

(b) Ethers

(c) Esters ✓

(d) Carboxylic acid

(x) On combustion the hydrocarbons produce

(a) CO_2 only

(b) H_2O only

(c) Both CO_2 and H_2O ✓

(d) Oxygen

C. Brief / short questions.

(i) Define organic chemistry.

Organic chemistry is the branch of chemistry that deals with the study of carbon compounds, especially hydrocarbons and their derivatives. It covers how these compounds are made, their structures, their properties and their reactions.

(ii) What do you understand by the term "Catenation"?

Catenation is the ability of carbon atoms to link with one another by covalent bonds to form long chains, branched chains and rings. It is the reason carbon can form millions of

compounds, from simple methane to large proteins and plastics.

(iii) What is a homologous series?

A homologous series is a family of organic compounds with the same functional group and similar chemical properties, in which each member differs from the next by a -CH2- group. For example, the alkanes methane, ethane, propane and butane form a homologous series.

(iv) Write the names and uses of different types of coals.

The types of coal are peat, lignite, bituminous coal and anthracite. Peat is the lowest grade, used locally as a poor fuel. Lignite, brown coal, is used in power stations. Bituminous coal is the common household and industrial fuel and is used to make coke. Anthracite is the highest grade, with about 80% carbon, and gives the most heat with the least smoke.

(v) Comment that the plants are main source of organic compounds.

Plants are the main source of organic compounds because they make them from carbon dioxide and water by photosynthesis, using sunlight. The sugars, starch, cellulose, oils and proteins that plants produce become the food, fibres, fuels and raw materials that animals and humans use, so almost all organic compounds trace back to plants.

(vi) How organic chemistry is important to study?

Organic chemistry is important because the substances of life itself, such as proteins, carbohydrates, fats and DNA, are organic compounds. Its study gives us medicines, plastics, fuels, dyes, soaps, fibres and pesticides, and it explains digestion, fermentation and the chemistry of the human body.

(vii) Define the functional group.

A functional group is a specific group of atoms in an organic molecule that determines the characteristic chemical properties and reactions of the compound. For example, the -OH group makes a compound an alcohol, and the -COOH group makes it a carboxylic acid, whatever the rest of the molecule is.

(viii) Draw structures of the isomers of a hydrocarbon having 5 carbon atoms?

The hydrocarbon C5H12 has three isomers, shown here in condensed form: n-pentane, CH3-CH2-CH2-CH2-CH3 (a straight chain); isopentane (2-methylbutane), CH3-CH(CH3)-CH2-CH3 (one branch); and neopentane (2,2-dimethylpropane), C(CH3)4 (two branches on the middle carbon). All three have the same formula but different structures and different boiling points.

(ix) What is an "alkyl group"?

An alkyl group is the group formed when one hydrogen atom is removed from an alkane molecule. It is represented by the symbol R, for example methyl, -CH3, from methane, and ethyl, -C2H5, from ethane. Alkyl groups attach to functional groups to build larger organic molecules.

(x) Why peat is inferior quality coal?

Peat is an inferior quality coal because it is the first, least mature stage of coal formation: it

contains a large amount of water and only a low percentage of carbon, about 50-60%. It therefore gives little heat on burning, produces much smoke and ash, and crumbles easily, unlike anthracite, which is about 80% carbon.

D. Reasoning and descriptive questions.

(i) What are the main differences between organic and Inorganic compounds?

Organic compounds contain carbon, usually with hydrogen, and are mostly covalent, with low melting and boiling points, slow reactions and solubility in organic solvents. Inorganic compounds are generally made of other elements, are often ionic, with high melting and boiling points, fast ionic reactions and solubility in water. Organic compounds burn to give CO2 and water, while most inorganic compounds do not burn.

(ii) Why hydrocarbons do not dissolve in water?

Hydrocarbons do not dissolve in water because they are non-polar molecules, while water is a polar solvent. Water molecules attract one another strongly through hydrogen bonding and will not separate to make room for non-polar hydrocarbon molecules, so the hydrocarbon stays as a separate layer. Like dissolves like: hydrocarbons dissolve instead in non-polar organic solvents.

(iii) Why writing structural formulae of the organic compounds is important instead of the molecular formulae?

Structural formulae are important because the molecular formula alone does not show how the atoms are joined, and compounds with the same molecular formula can have different structures and properties. For example, C4H10 can be n-butane or isobutane, which have different boiling points. The structural formula shows the arrangement of atoms and the functional group, which decides the chemistry of the compound.

(iv) Does catenation is shown by the carbon element only?

No, catenation is not shown by carbon alone, but carbon shows it to the greatest extent. Other elements such as sulphur, silicon and phosphorus can also form chains, but their chains are short and unstable, while carbon forms very strong C-C bonds and can build chains, branches and rings of thousands of atoms, which is why organic chemistry is so large.

(v) In the LPG, the fraction of propane and butane is more than the Methane gas.

This is so because LPG is made to be stored as a liquid under moderate pressure in cylinders. Propane and butane liquefy easily under pressure at ordinary temperature, while methane, the main part of natural gas, stays a gas and needs very high pressure or very low temperature to liquefy. So LPG is rich in propane and butane, and methane is supplied separately as natural gas or CNG.

(vi) Discuss the organic chemistry evolution. How Wohler broke the concept of vitalism?

Early chemists believed in vitalism: that organic compounds could only be made by living

things with a mysterious vital force, and never in the laboratory. In 1828 Friedrich Wohler heated ammonium cyanate, an inorganic salt, and obtained urea, a compound found in animal urine. This proved that an organic compound could be made from inorganic material without any living organism, and the vital force theory collapsed. Organic chemistry then grew into the study of carbon compounds, natural and synthetic.

(vii) How the organic chemistry helps the mankind?

Organic chemistry helps mankind by providing the materials of daily life: fuels such as petrol, diesel and natural gas; medicines such as aspirin, antibiotics and antiseptics; fibres and plastics for clothes, houses and machines; soaps, detergents, dyes and cosmetics; and fertilizers and pesticides that increase food production. It also explains the chemistry of food, digestion and the human body.

(viii) What are the general characteristics of the organic compounds?

Organic compounds are mostly covalent, with low melting and boiling points, and they are usually insoluble in water but soluble in organic solvents. They burn in air to give carbon dioxide and water, they react slowly because covalent bonds must be broken, and they show isomerism, so many compounds share the same molecular formula. They are generally poor conductors of electricity.

(ix) How crude oil is purified in the oil refineries?

Crude oil is purified by fractional distillation in a refinery. The oil is heated to vapour and passed into a tall fractionating column, which is hot at the bottom and cooler at the top. Fractions condense at different levels according to their boiling points: refinery gases at the top, then gasoline, kerosene, diesel, lubricating oil and wax, with bitumen left at the bottom. Each fraction is then treated further to remove impurities.

(x) Write a note on classification of alkane hydrocarbons?

Alkanes are classified by the shape of their carbon chain into straight-chain (normal) alkanes, such as n-butane, branched-chain alkanes, such as isobutane, and cyclic alkanes (cycloalkanes), such as cyclohexane. They are also classified by the type of carbon atom they contain: primary carbons joined to one other carbon, secondary carbons joined to two, tertiary to three, and quaternary to four. All alkanes share the general formula CnH2n+2 and are saturated hydrocarbons.

(xi) What are the sources of the organic compounds?

The sources of organic compounds are natural and synthetic. Natural sources are plants, which make sugars, starch, oils and proteins; animals, which give fats, proteins and urea; and fossil sources, coal, petroleum and natural gas, formed from ancient plants and animals. Synthetic sources are the chemical industries, which make plastics, medicines, dyes, fibres and fertilizers from coal and petroleum products.

(xii) What are alkanes? How they are important to us?

Alkanes are saturated hydrocarbons with only single bonds between carbon atoms, with the

general formula CnH2n+2. They are important as fuels: methane is the main part of natural gas, LPG contains propane and butane, petrol and diesel are mixtures of higher alkanes, and kerosene is used for cooking and lighting. They are also used as solvents and as raw materials for plastics and other chemicals.

(xiii) Discuss the importance of the functional groups? What are the different types of functional groups?

Functional groups are important because they decide the chemical behaviour of an organic compound: compounds with the same functional group react in the same way, whatever the rest of the molecule. The main functional groups are hydroxyl (-OH) in alcohols, carboxyl (-COOH) in carboxylic acids, carbonyl (C=O) in aldehydes and ketones, amino (-NH2) in amines, alkoxy (-O-) in ethers, ester (-COO-) in esters, and carbon-carbon double and triple bonds in alkenes and alkynes.

(xiv) What is the condensed formula used for the organic compound? Draw the condensed formula of first ten straight chain alkanes with their names.

A condensed formula is a short way of writing the structure of an organic compound in one line, showing the groups in order without drawing every bond. The first ten straight-chain alkanes are: methane CH4, ethane CH3CH3, propane CH3CH2CH3, butane CH3(CH2)2CH3, pentane CH3(CH2)3CH3, hexane CH3(CH2)4CH3, heptane CH3(CH2)5CH3, octane CH3(CH2)6CH3, nonane CH3(CH2)7CH3, and decane CH3(CH2)8CH3.

Chapter 12: Hydrocarbons

Exercise (Chapter 12)

A. Fill in the blanks. (Filling word shown in red.)

(i) The hydrocarbons are the compound containing **carbon** and **hydrogen** elements only.

(ii) Saturated hydrocarbons are relatively **less** in reactivity.

(iii) Unsaturated hydrocarbons are **more** reactive.

(iv) Alkenes react with halogens at **room** temperature.

(v) The water solution of KMnO4 forms **glycols** with alkenes.

(vi) Alkenes are **less** reactive than Alkynes.

(vii) Benzene is an **aromatic** hydrocarbon.

(viii) Acetylene gas is used in **welding**.

(ix) Alkanes are also known as **paraffins**.

(x) Hydrogenation process results in formation of **ghee** from oils.

B. Choose the correct answer. (Correct option marked with a green tick.)

(i) The simplest member of hydrocarbon family is the

(a) C4H10

(b) C3H8

(c) C2H6

(d) CH4 ✓

(ii) The Alkynes have the general formula

(a) CnH2n+2

(b) CnH2n+0

(c) CnH2n-2 ✓

(d) CnH2n+4

(iii) Dehydration of alcohols with sulphuric acid yields:

(a) Alkene ✓

(b) Alkyne

(c) Alkane

(d) Benzene

(iv) Glycols are the compounds having:

(a) adjacent halogens

(b) adjacent hydroxyl groups ✓

(c) alternate hydroxyl groups

(d) alternate halogens

(v) To prepare ghee the vegetable oil is reacted with:

(a) hydrogen ✓

(b) halogen

(c) oxygen

(d) nitrogen

(vi) The secondary carbon atoms is linked with the number of carbon atoms

(a) 1

(b) 2 ✓

(c) 3

(d) 4

(vii) Hydrogenation reaction utilizes catalyst:

(a) Ni or Pd ✓

(b) Zn

(c) H2

(d) do not requires any catalyst

(viii) General formula of Alkanes is

(a) CnH2n-2

(b) CnH2n+2 ✓

(c) CnH2n

(d) CnH2n-1

(ix) Halogenation means introduction of

(a) Cl

(b) Br

(c) I

(d) All of these ✓

(x) Olefin is another name for

(a) Alkanes

(b) Alkenes ✓

(c) Alkynes

(d) Aromatic compounds

C. Brief / short questions.

(i) Define hydrocarbons.

Hydrocarbons are organic compounds that contain only carbon and hydrogen. They are the simplest organic compounds, and include alkanes, alkenes, alkynes and aromatic compounds such as benzene.

(ii) What is the difference between alicyclic and aromatic hydrocarbons?

Alicyclic hydrocarbons are ring compounds that behave like open-chain aliphatic compounds, for example cyclohexane, which has only single bonds and no special stability. Aromatic hydrocarbons contain a benzene ring with delocalized pi electrons, for example benzene itself, which gives them unusual stability and a different pattern of reactions,

mainly substitution instead of addition.

(iii) Explain what is the difference between saturated and unsaturated compounds?

Saturated compounds contain only single bonds between carbon atoms, so each carbon holds the maximum possible number of hydrogen atoms, as in alkanes. Unsaturated compounds contain at least one double or triple bond, so their carbons hold fewer hydrogens, as in alkenes and alkynes. Unsaturated compounds are more reactive because their multiple bonds can break open in addition reactions.

(iv) Why chlorine and methane react in light?

Chlorine and methane react in light because light energy breaks the Cl-Cl bond into two highly reactive chlorine free radicals, which start a chain reaction. A chlorine radical removes a hydrogen from methane to form HCl and a methyl radical, which then reacts with another chlorine molecule, and the chain continues. Without light, the first radicals are not formed and the reaction does not start.

(v) Describe the chemistry of combustion.

Combustion is the burning of a substance in oxygen, a rapid exothermic reaction. A hydrocarbon burns completely in a good supply of oxygen to give carbon dioxide and water with release of heat, for example $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$. In a poor supply of oxygen the combustion is incomplete and gives carbon monoxide and carbon (soot) instead, which is poisonous and wasteful.

(vi) What do you understand by the term "homologous series"?

A homologous series is a family of organic compounds with the same functional group and similar chemical properties, in which each member differs from the next by a -CH_2- group. The alkanes, alkenes and alcohols are each a homologous series, and within a series the physical properties change gradually with molecular size.

(vii) Describe the difference between dehydrogenation and dehydrohalogenation?

Dehydrogenation is the removal of hydrogen from a compound, for example an alkane losing hydrogen to form an alkene, usually with heat and a catalyst. Dehydrohalogenation is the removal of a hydrogen atom and a halogen atom from adjacent carbons of a haloalkane, usually with alcoholic KOH, to form an alkene and a hydrogen halide. Both produce an alkene, but one removes H_2 and the other removes HX.

(viii) What is meant by dehydration? What products are formed by dehydration of alcohols?

Dehydration is the removal of water from a compound. When an alcohol is dehydrated, usually by heating with concentrated sulphuric acid, it loses a water molecule from neighbouring carbons and forms an alkene: for example, ethanol gives ethene, $C_2H_5OH \rightarrow C_2H_4 + H_2O$.

(ix) What is the role of acetylene gas in fruit ripening?

Acetylene gas is used to ripen fruits artificially because it acts like the natural plant hormone ethylene, which triggers ripening. Fruit dealers expose unripe fruit to acetylene (from calcium carbide and water) so that it ripens faster, softens and changes colour. Its use is discouraged in many places because impure acetylene can carry harmful traces of arsenic and phosphorus compounds.

(x) Write chemical test to distinguish between saturated and unsaturated organic compounds?

The bromine water test distinguishes them: add a few drops of brown bromine water to each compound. An unsaturated compound decolourises the bromine water quickly, because bromine adds across its double or triple bond, while a saturated compound leaves the brown colour unchanged. The Baeyer test with dilute purple KMnO4 gives the same distinction: the purple colour disappears with an unsaturated compound.

D. Reasoning and descriptive questions.

(i) Why alkenes are more reactive than alkanes?

Alkenes are more reactive than alkanes because they contain a carbon-carbon double bond, whose pi bond is weaker and more exposed than a sigma bond. The pi electrons are easily attacked by reagents, so the double bond breaks open in addition reactions. Alkanes have only strong single sigma bonds, so they react only by substitution under harsh conditions.

(ii) Why hydrocarbons do not dissolve in water?

Hydrocarbons do not dissolve in water because they are non-polar, while water is polar and its molecules are held together by strong hydrogen bonds. Water molecules prefer to stay bonded to each other rather than separate to surround non-polar hydrocarbon molecules, so the hydrocarbon forms a separate layer. Hydrocarbons dissolve in non-polar organic solvents instead.

(iii) In your opinion, which is more reactive, Acetylene or Ethene? And why?

Acetylene (ethyne) is more reactive than ethene. It has a triple bond with two pi bonds, so it has more exposed pi electron density available for attack, and it undergoes addition reactions more readily, for example it adds bromine twice. Ethyne is also acidic, since its hydrogen can be removed by strong bases to form acetylides, which ethene cannot do.

(iv) n-pentane boils at higher temperature than n-propane?

n-Pentane boils at a higher temperature than n-propane because it is a larger molecule with a greater surface area, so the van der Waals forces between its molecules are stronger. More heat energy is needed to overcome these forces, so pentane boils at 36 °C while propane, a smaller molecule with weaker forces, boils at -42 °C.

(v) Alkenes form addition products where as alkane form substituted products?

Alkenes form addition products because their double bond can break open, allowing atoms

to add across the two carbons without losing any atom from the molecule, for example ethene + Br2 → 1,2-dibromoethane. Alkanes have no multiple bond to open, so a reagent can only react by replacing a hydrogen atom, giving a substitution product, for example methane + Cl2 → chloromethane + HCl in sunlight.

(vi) What do you understand by the term 'hydrocarbon'? how they have been produced in nature?

Hydrocarbons are compounds of carbon and hydrogen only. In nature they were produced from the remains of plants and animals buried under the earth millions of years ago: under heat, pressure and bacterial action, the organic matter changed slowly into coal, petroleum and natural gas, which are the natural sources of hydrocarbons.

(vii) How the hydrocarbons are classified? Explain with examples.

Hydrocarbons are classified first into open-chain (aliphatic) and closed-chain (cyclic) compounds. Open-chain hydrocarbons are alkanes (single bonds, e.g. methane), alkenes (double bond, e.g. ethene) and alkynes (triple bond, e.g. ethyne). Cyclic hydrocarbons are alicyclic (rings with single or multiple bonds, e.g. cyclohexane) and aromatic (benzene ring, e.g. benzene, toluene).

(viii) Write a comprehensive note on chemistry of alkanes.

Alkanes are saturated hydrocarbons, CnH2n+2, with only single bonds. They are generally unreactive: they burn in oxygen to give CO2 and water, they undergo free-radical substitution with halogens in sunlight, for example chlorination of methane, and they crack at high temperature into smaller alkanes and alkenes. They are used as fuels (natural gas, petrol, diesel, kerosene) and as solvents and raw materials.

(ix) How you can prepare the alkenes? Describe the reactivity of alkenes too.

Alkenes are prepared by dehydration of alcohols with concentrated sulphuric acid, by dehydrohalogenation of haloalkanes with alcoholic KOH, and by cracking of larger alkanes. Alkenes are reactive because of their double bond: they undergo addition reactions with hydrogen (hydrogenation), halogens, hydrogen halides and water, they decolourise bromine water and KMnO4, and they polymerise to form plastics such as polythene.

(x) Explain various methods for the preparation of alkynes?

Alkynes are prepared by reacting calcium carbide with water to give acetylene, CaC2 + 2H2O → C2H2 + Ca(OH)2, by dehydrohalogenation of vicinal dihalides with alcoholic KOH, for example 1,2-dibromoethane giving ethyne, and by reacting a terminal alkyne salt with an alkyl halide to build longer alkynes.

(xi) Explain the chemical properties of alkynes?

Alkynes are reactive because of their triple bond: they undergo addition of hydrogen, halogens and hydrogen halides, in two stages, first to an alkene and then to an alkane derivative. They burn with a very hot, smoky flame and are used in oxy-acetylene welding. Terminal alkynes are acidic and react with ammoniacal silver nitrate or cuprous chloride to

give metal acetylides, a test that distinguishes them from alkenes.

(xii) State the general rules for naming of alkanes by giving examples.

The rules for naming alkanes are: choose the longest continuous carbon chain as the parent, and name it methane, ethane, propane and so on; number the chain from the end that gives the branches the lowest numbers; name the branches as methyl, ethyl, etc., with their position numbers; and list branches alphabetically if there are several. For example, CH3-CH(CH3)-CH2-CH3 is 2-methylbutane, not 3-methylbutane, because numbering starts from the nearer end.

(xiii) Draw the structure of following hydrocarbons? i) 3,3-Dimethyl pentane ii) 2-Methyl hexane iii) n-Hexane iv) 3,4-Dimethyl heptane v) Isopentane. vi) Cyclo hexane

The structures in condensed form are: i) 3,3-dimethylpentane: CH3CH2C(CH3)2CH2CH3. ii) 2-methylhexane: CH3CH(CH3)CH2CH2CH2CH3. iii) n-hexane: CH3CH2CH2CH2CH2CH3. iv) 3,4-dimethylheptane: CH3CH2CH(CH3)CH(CH3)CH2CH2CH3. v) Isopentane (2-methylbutane): CH3CH(CH3)CH2CH3. vi) Cyclohexane: a six-carbon ring, (CH2)6 in a closed ring, each carbon joined to two neighbours and two hydrogens.

Chapter 13: Biochemistry

Exercise (Chapter 13)

A. Fill in the blanks. (Filling word shown in red.)

(i) Biochemistry is the study of compounds which are found in **living organisms**.

(ii) Vitamin-D is a **fat** soluble vitamin.

(iii) Carbohydrates are classified according to the number of **saccharide** units.

(iv) Lactose is a **disaccharide** sugar.

(v) There are **10** essential amino acids.

(vi) Glyceraldehyde is a **three** carbon sugar.

(vii) Sucrose is also known as **table sugar**.

(viii) Proteins are composed of **amino acids**.

(ix) There are **two** types of nucleic acids.

(x) Deficiency of vitamin-C causes **scurvy**.

B. Choose the correct answer. (Correct option marked with a green tick.)

(i) Plants mainly consist of:

(a) Carbohydrates ✓

(b) Proteins

(c) Lipids

(d) Nucleic acids

(ii) Monosaccharide has the formula:

(a) $C_nH_{2n}O_2$

(b) $C_6H_{12}O_6$

(c) $C_n(H_2O)_{n-1}$

(d) $C_n(H_2O)_n$ ✓

(iii) Monosaccharides are composed of:

(a) 1 glucose unit ✓

(b) 2 glucose units

(c) 3 glucose units

(d) 4 glucose units

(iv) The amino acids contain the groups:

(a) Amino group only

(b) Carboxyl group only

(c) Hydroxyl group only

(d) Both amino and carboxyl group ✓

(v) Lipids are the:

(a) Simple esters of fatty acids

(b) Long chained esters of fatty acids ✓

(c) Short chained esters of fatty acids

(d) Saturated fatty acids only

(vi) The number of essential amino acids in the body is:

(a) 10 ✓

(b) 20

(c) 30

(d) 40

(vii) DNA molecule is concerned with:

(a) Energy

(b) Heredity characters ✓

(c) Respiration

(d) Fats

(viii) Fats:

(a) Provide energy

(b) Make insulation

(c) Act as cushion

(d) Perform all these functions ✓

(ix) Deficiency of vitamin-A causes:

(a) Night blindness ✓

(b) Beriberi

(c) Scurvy

(d) Rickets

(x) Vitamin-D can be obtained:

(a) From balanced diet only

(b) From medicines only

(c) When body is exposed to sunlight ✓

(d) From balanced diet only

C. Brief / short questions.

(i) Comment that the carbohydrates are hydrates of carbon?

Carbohydrates are called hydrates of carbon because most of them have the general formula Cx(H2O)y, that is, carbon combined with hydrogen and oxygen in the same ratio as in water, 2:1. For example, glucose is C6H12O6, which can be written as C6(H2O)6. The name is not exact, because some carbohydrates, such as deoxyribose, do not fit the formula, but it describes most of them.

(ii) Name the functional groups present in the carbohydrates?

Carbohydrates contain hydroxyl groups (-OH) on nearly every carbon, and one carbonyl group: an aldehyde group (-CHO) in aldoses such as glucose, or a ketone group (C=O) in

ketoses such as fructose. These groups give carbohydrates their sweet taste, their solubility in water and their reducing properties.

(iii) Do nucleic acids contain a sugar unit? If yes, which sugar they contain?

Yes, nucleic acids contain a sugar unit in every nucleotide. DNA contains the five-carbon sugar deoxyribose, while RNA contains ribose. The sugar joins to a phosphate group and a nitrogenous base to form the nucleotide, and the nucleotides link into the long chains of DNA and RNA.

(iv) Write the functions of oligosaccharides?

Oligosaccharides, made of 2 to 10 sugar units, serve as a quick source of energy, for example sucrose and lactose, and as storage and transport sugars in plants and animals. They also take part in cell recognition: oligosaccharides attached to proteins and lipids on cell surfaces help cells recognize each other and take part in the immune response.

(v) What functional groups are present in the amino acids?

Every amino acid contains an amino group (-NH2) and a carboxyl group (-COOH) attached to the same central carbon atom, along with a hydrogen atom and a side chain (R group) that differs from one amino acid to another. It is the side chain that gives each amino acid its own character, acidic, basic or neutral.

(vi) Write a note on essential amino acids?

Essential amino acids are the amino acids that the human body cannot make itself, so they must be obtained from food. The book counts 10 of them. They are needed to build proteins, enzymes and hormones, and a diet lacking any one of them causes poor growth and illness. Good sources are meat, eggs, milk and pulses.

(vii) What are the fatty acids?

Fatty acids are long-chain carboxylic acids, usually with 4 to 24 carbon atoms, that are the building blocks of fats and oils. They may be saturated, with only single bonds, as in stearic acid, or unsaturated, with one or more double bonds, as in oleic acid. Fats and oils are esters formed when fatty acids combine with glycerol.

(viii) What are the biological roles of steroids?

Steroids are lipids with a four-ring structure that play key roles in the body: cholesterol keeps cell membranes flexible, bile salts help digest fats, and steroid hormones such as testosterone, oestrogen and cortisol control growth, reproduction and the response to stress. Vitamin D is also a steroid that controls calcium in bones.

(ix) What are the vitamins? How they are classified?

Vitamins are organic compounds needed in very small amounts for normal growth, health and metabolism, which the body cannot make in enough quantity. They are classified by solubility: fat-soluble vitamins A, D, E and K, which are stored in body fat and the liver, and water-soluble vitamins, the B group and vitamin C, which are not stored and must be taken daily in food.

(x) How vitamin-D is produced in the body?

Vitamin D is produced in the skin when it is exposed to sunlight: ultraviolet rays convert a cholesterol-like substance in the skin, 7-dehydrocholesterol, into vitamin D3. This travels to the liver and kidneys, where it is changed into its active form, which helps the body absorb calcium and phosphorus for strong bones and teeth.

D. Reasoning and descriptive questions.

(i) Lipids provide more energy than carbohydrates. Why?

Lipids provide more energy than carbohydrates because they contain much less oxygen and far more carbon-hydrogen bonds per gram. When these C-H bonds are oxidized, they release more energy: fats give about 9 kcal per gram, while carbohydrates give only about 4 kcal per gram. That is why the body stores its long-term energy as fat.

(ii) Why carbohydrates constitute the plant body, whereas protein constitutes the animal body?

Plants build their bodies mainly from carbohydrates because they make sugars by photosynthesis and use cellulose, a carbohydrate, to build their cell walls and give the plant its structure. Animals do not photosynthesize; they build their bodies from proteins, which form muscles, skin, enzymes, hormones and blood, and so protein is the main structural material of the animal body.

(iii) Why the cellulose is not digested by human beings, while cows digest it?

Cellulose is not digested by humans because the human gut does not produce cellulase, the enzyme that breaks the beta-glycosidic bonds of cellulose, so it passes through as fibre. Cows digest it because their rumen contains bacteria and protozoa that produce cellulase, which breaks cellulose down into sugars the cow can absorb.

(iv) The proteins of beef and beans are same? If not, why?

The proteins of beef and beans are not the same, because proteins differ in the kinds and order of their amino acids. Beef is an animal protein containing all the essential amino acids in the right proportions, so it is a complete protein. Bean protein is a plant protein that is short of some essential amino acids, such as methionine, so it is an incomplete protein, although beans plus cereals together give a complete supply.

(v) Why water-soluble vitamins are required daily, while fat-soluble are not?

Water-soluble vitamins, the B group and vitamin C, dissolve in water, are not stored in the body in any large amount, and the excess is excreted in urine, so they must be taken in food every day. Fat-soluble vitamins A, D, E and K dissolve in fat and are stored in the liver and body fat, so the body can draw on its store and they are not needed daily.

(vi) What are the carbohydrates? Discuss the classification of carbohydrates and also their importance?

Carbohydrates are organic compounds of carbon, hydrogen and oxygen, usually with the formula Cx(H2O)y, found mainly in plants. They are classified as monosaccharides (one

sugar unit, e.g. glucose, fructose), disaccharides (two units, e.g. sucrose, lactose, maltose) and polysaccharides (many units, e.g. starch, cellulose, glycogen). They are important as the main source of energy for the body, as storage food (starch in plants, glycogen in animals) and as structural material (cellulose in plant walls).

(vii) Write the chemistry of the proteins.

Proteins are large molecules made of amino acids joined by peptide bonds: the carboxyl group of one amino acid links to the amino group of the next, releasing water. Chains of amino acids fold into specific shapes, primary, secondary, tertiary and quaternary structures, which decide the protein function. Proteins act as enzymes, hormones, antibodies, and structural materials such as muscle, skin and hair, and on hydrolysis they break back into amino acids.

(viii) What are lipids? Also discuss the importance of lipids.

Lipids are organic compounds insoluble in water but soluble in organic solvents, including fats, oils, waxes, steroids and phospholipids. They are important as concentrated energy stores, giving 9 kcal per gram; as insulation under the skin and around organs; as the main material of cell membranes (phospholipids); and as hormones and vitamins, such as steroid hormones and vitamin D.

(ix) What are the vitamins? Discuss their effect on the life of human beings.

Vitamins are organic substances needed in tiny amounts to keep the body healthy, although they give no energy themselves. They act as coenzymes in metabolism, build bones and blood, protect sight, skin and nerves, and prevent deficiency diseases: lack of vitamin A causes night blindness, vitamin B1 causes beriberi, vitamin C causes scurvy, and vitamin D causes rickets. A balanced diet supplies them all.

(x) Write a note on the nature and role of nucleic acids.

Nucleic acids, DNA and RNA, are long chains of nucleotides, each made of a sugar, a phosphate and a nitrogenous base. DNA, deoxyribonucleic acid, is a double helix that stores the hereditary information of the cell and passes it to the next generation. RNA, ribonucleic acid, copies that information and carries it to the ribosomes, where it directs the building of proteins. Together they control growth, reproduction and the working of the cell.

(xi) Name the diseases caused by the deficiency of vitamins.

Vitamin A deficiency causes night blindness and dry skin. Vitamin B1 deficiency causes beriberi, and B3 deficiency causes pellagra. Vitamin C deficiency causes scurvy, with bleeding gums and weak joints. Vitamin D deficiency causes rickets in children and osteomalacia in adults, with soft, weak bones. Lack of vitamin K causes slow clotting of blood, and lack of vitamin B12 causes anaemia.

(xii) What are the important sources of vitamins?

The important sources of vitamins are: vitamin A from carrots, butter, eggs and fish liver oil; B vitamins from whole grains, meat, eggs and pulses; vitamin C from citrus fruits,

tomatoes, guava and green vegetables; vitamin D from sunlight, fish liver oil, eggs and milk; vitamin E from vegetable oils, nuts and seeds; and vitamin K from green leafy vegetables. A balanced diet with fruits, vegetables, milk and cereals supplies all of them.

Chapter 14: Environmental Chemistry-I: The Atmosphere

Exercise (Chapter 14)

A. Fill in the blanks. (Filling word shown in red.)

(i) The **amount** of oxygen, nitrogen, and carbon dioxide in the mesosphere is almost same as that near the earth.

(ii) The **highest** temperature in the atmosphere is found in thermosphere.

(iii) The biodegradable pollutants decompose by the help of **micro-organisms**.

(iv) Ozone layer prevents the **ultraviolet radiations** coming from the Sun.

(v) The main cause of ozone depletion are **CFCs** (chlorofluorocarbons).

(vi) Acid rain has pH **less than 5.6**.

(vii) Smog is the mixture of **smoke and fog**.

(viii) The main cause of acid rain is the oxides of **sulphur and nitrogen**.

(ix) The temperature of mesosphere ranges from **20 °C to -80 °C**.

(x) The oxygen present in atmosphere is **21%**.

B. Choose the correct answer. (Correct option marked with a green tick.)

(i) The ratio of oxygen gas present in air is:

(a) 78%

(b) 21% ✓

(c) 0.93%

(d) 0.04%

(ii) The Troposphere is the:

(a) Uppermost level

(b) Lowermost level ✓

(c) Second level

(d) Third level

(iii) Which of the following is a primary pollutant?

(a) SO_2

(b) CFCs

(c) CO

(d) All of the above ✓

(iv) The main cause of acid rain is:

(a) SO_2 only

(b) NO_2 only

(c) Both SO_2 and NO_2 ✓

(d) CO_2 only

(v) The greenhouse effect results in:

(a) Increase in temperature ✓

(b) Decrease in temperature

(c) Increase in rainfall

(d) Decrease in rainfall

(vi) Plastic is an example of:

(a) Biodegradable pollutant

(b) Non-biodegradable pollutant ✓

(c) Primary pollutant

(d) Secondary pollutant

(vii) Thermosphere lasts up to:

(a) 100-350 km ✓

(b) 50-100 km

(c) 10-50 km

(d) 350-500 km

(viii) Ozone layer prevents:

(a) Infrared rays

(b) Visible light

(c) Ultraviolet rays from sun and stars ✓

(d) X-rays

(ix) The ratio of carbon dioxide gas present in air is:

(a) 0.04% ✓

(b) 0.93%

(c) 21%

(d) 78%

(x) In thermosphere the ratio of O_2 is:

(a) 21%

(b) 78%

(c) 0% ✓

(d) 0.04%

C. Reasoning and descriptive questions.

(i) Smog is mostly present in the winter season. Why?

Smog is mostly present in winter because cold air is heavier and stays near the ground, trapping smoke and fog below it in a temperature inversion, while there is little wind to disperse them. In winter people also burn more coal and wood for heating, adding more smoke, and the weak sunlight cannot break the smog up, so it hangs over cities for days.

(ii) Ozone is a pollutant but it is necessary for us. Why?

Ozone is a pollutant at ground level because it irritates the eyes, lungs and throat and damages crops, and it forms in photochemical smog. But ozone high in the stratosphere is necessary because it absorbs most of the harmful ultraviolet radiation from the sun before it reaches the earth, protecting living things from skin cancer, eye damage and damage to crops and plankton.

(iii) Carbon dioxide is a pollutant but it is essential for us. Why?

Carbon dioxide is called a pollutant when it is present in excess, because it traps heat and causes global warming. But it is essential because plants need it for photosynthesis, which produces the oxygen we breathe and the food we eat. Without carbon dioxide in the air, plants could not grow and nearly all life on earth would stop.

(iv) What measures can be taken to fill the ozone hole?

The ozone hole can be filled by stopping the release of ozone-destroying chemicals, especially CFCs from old refrigerators, air conditioners and spray cans, as required by the Montreal Protocol. Using CFC-free alternatives, recycling old equipment so its gases do not escape, and planting trees, which absorb carbon dioxide and cool the air, all help the ozone layer recover, which scientists expect to take several decades.

(v) How global warming is caused?

Global warming is caused by the greenhouse effect: gases such as carbon dioxide, methane, nitrous oxide and CFCs, released by burning fossil fuels, deforestation and industry, trap heat radiating from the earth instead of letting it escape to space. As these gases build up, the average temperature of the earth rises, melting ice, raising sea levels and changing rainfall patterns.

(vi) How the atmosphere is sustaining life on the earth?

The atmosphere sustains life by providing the gases living things need: oxygen for respiration, carbon dioxide for photosynthesis and nitrogen for proteins. It keeps the earth warm through the greenhouse effect, shields life from harmful ultraviolet radiation through the ozone layer, burns up most meteors, and carries the water cycle that brings rain to the land.

(vii) Describe the composition of earth atmosphere?

The atmosphere is a mixture of gases: about 78% nitrogen, 21% oxygen, 0.93% argon and 0.04% carbon dioxide, with small amounts of neon, helium, methane and ozone, plus variable water vapour and dust. Nitrogen and oxygen make up about 99% of dry air, and the composition stays almost the same up to about 80 km height.

(viii) What is the pollution? How the pollution is created?

Pollution is the addition of harmful substances, or pollutants, to the environment in amounts that damage living things and their surroundings. It is created by human activities: burning fossil fuels releases smoke and gases into the air, factories and sewage discharge wastes into water, and farming and households add chemicals and solid waste to

soil and water, disturbing the natural balance.

(ix) What are pollutants? What is the difference between primary and secondary pollutants?

Pollutants are substances that pollute the environment, such as smoke, sulphur dioxide, carbon monoxide, CFCs and plastic waste. Primary pollutants are released directly from a source, for example SO2 and CO from chimneys and exhausts. Secondary pollutants are not released directly but form in the air by reactions between primary pollutants, for example ozone and peroxyacetyl nitrate formed in photochemical smog.

(x) Describe the effect of ozone layer on the life of earth? What are the main causes of ozone depletion?

The ozone layer protects life by absorbing most of the sun harmful ultraviolet rays, which would otherwise cause skin cancer, cataracts, damage to crops and death of plankton in the oceans. The main causes of ozone depletion are chlorofluorocarbons (CFCs) from refrigerators, air conditioners and sprays, which release chlorine atoms that destroy ozone, along with halons, nitrogen oxides from aircraft and methyl bromide from fumigation.

(xi) How ozone is produced in the atmosphere?

Ozone is produced in the stratosphere when ultraviolet light from the sun splits an oxygen molecule, O2, into two oxygen atoms. Each free oxygen atom then joins another oxygen molecule to form ozone, O3. Ozone itself absorbs ultraviolet light and breaks back into O2 and an oxygen atom, so a natural balance of formation and destruction keeps the ozone layer steady.

Chapter 15: Environmental Chemistry-II: Water

Exercise (Chapter 15)

A. Fill in the blanks. (Filling word shown in red.)

(i) Water covers **71**% of the earth surface.

(ii) Fresh water is only **2.5**% of the total water.

(iii) Pure water has **no** smell.

(iv) Water dissolves most of the **inorganic** and many **organic** compounds.

(v) Antarctic ice sheet contains **68.7**% of the total fresh water reservoir.

(vi) Only a small fraction of the total water is available for us, which is about **0.7**%.

(vii) Water pollution is caused due to dissolved **heavy metals** and **organic toxins**.

(viii) Gastro is caused by **contaminated water**.

(ix) Jaundice is a disease in which the body becomes **yellow**.

(x) High fluoride in water causes **fluorosis**.

B. Choose the correct answer. (Correct option marked with a green tick.)

(i) Pure water is:

(a) Sour in taste

(b) Sweet in taste

(c) Bitter in taste

(d) Tasteless ✓

(ii) The bicarbonates and sulphates responsible for hardness of water are of:

(a) Ca only

(b) Mg only

(c) Both of these ✓

(d) None of these

(iii) The major responsibility for polluting the water rests with:

(a) Humans ✓

(b) Animals

(c) Plants

(d) Fishes

(iv) Hard water can be drunk safely:

(a) Always

(b) Never

(c) Depends upon the type of impurity ✓

(d) Only after boiling

(v) Hard water can be used for washing:

(a) Easily

(b) With difficulty ✓

(c) Not at all

(d) Only with soap

(vi) Water locked in polar ice caps is:

(a) 68.7% ✓

(b) 30.1%

(c) 1.2%

(d) 2.5%

(vii) The largest source of fresh water is:

(a) Rivers

(b) Lakes

(c) Ground water

(d) Polar ice caps ✓

(viii) The process of solvation makes:

(a) Solution ✓

(b) Suspension

(c) Colloid

(d) Precipitate

(ix) Hard water contains ions of:

(a) Sodium

(b) Potassium

(c) Iron

(d) Calcium or Magnesium ✓

(x) The major cause of water pollution is:

(a) Sewage of Industries ✓

(b) Rain water

(c) Sea water

(d) Ground water

C. Reasoning and descriptive questions.

(i) Rain water is always soft. Why?

Rain water is always soft because it forms by evaporation and condensation, and it has not yet flowed over rocks or soil to pick up dissolved calcium and magnesium salts. Hardness comes from dissolved bicarbonates and sulphates of calcium and magnesium, and rain water contains almost none of them, although it may pick up a little dust and dissolved gases as it falls.

(ii) Why the UV energy is used to make water drinkable?

Ultraviolet energy is used to make water drinkable because UV light kills or inactivates bacteria, viruses and other germs in the water by damaging their DNA, so they cannot

reproduce and cause disease. It adds no chemicals and no taste to the water, and it is a simple, cheap method for disinfecting drinking water.

(iii) Pure water is tasteless, then why bottled water has marked taste?

Pure water itself is tasteless, but bottled water has a marked taste because it contains dissolved minerals, such as calcium, magnesium and sodium salts, picked up from its spring or added during treatment. Different brands have different mineral contents, and the plastic bottle, storage and added carbon dioxide in sparkling water also change the taste.

(iv) Why the use of pesticides should not be encouraged?

The use of pesticides should not be encouraged because they wash from fields into rivers and groundwater, poisoning fish and other aquatic life, and they enter the food chain and harm birds, animals and humans. They also kill useful insects such as bees, and pests become resistant to them, so farmers need ever stronger doses, which increases the pollution.

(v) How malaria fever is associated with water?

Malaria is associated with water because the mosquito that carries the malaria parasite breeds in stagnant water, such as pools, ponds and water standing in pots and ditches. The female Anopheles mosquito lays its eggs in still water, the larvae grow there, and the adult mosquitoes then bite people and spread the parasite, so standing water near houses increases malaria.

(vi) Discuss the effects of water on environment?

Water shapes the environment in many ways: it supports all living things, carries nutrients through soil and rivers, and controls climate by storing heat and bringing rain. But too much water causes floods that destroy crops and houses, too little causes drought, and polluted water spreads disease and kills aquatic life. Water erosion also wears away soil and shapes valleys and coasts.

(vii) What is the difference between hard and soft water? Can hardness of water be removed?

Hard water contains dissolved calcium and magnesium salts, especially bicarbonates and sulphates, which stop soap from lathering and leave scale in kettles and pipes. Soft water contains little or none of these salts and lathers easily. Yes, hardness can be removed: temporary hardness is removed by boiling or Clark method with slaked lime, and permanent hardness is removed by ion exchange, washing soda or zeolite methods.

(viii) Why water dissolves most of the common compounds?

Water dissolves most common compounds because it is a polar molecule: its oxygen end is slightly negative and its hydrogen ends slightly positive, so it surrounds and pulls apart the ions or polar molecules of a solute. This process, called solvation, overcomes the forces holding the solute together, so most inorganic salts and many organic compounds dissolve in water, which is why water is called the universal solvent.

(ix) Write a note on water pollution?

Water pollution is the contamination of lakes, rivers, oceans and groundwater by substances that harm living things, mainly caused by humans. Its causes are industrial wastes containing heavy metals and toxic chemicals, household sewage and detergents, agricultural fertilizers and pesticides washed from fields, and oil spills. Polluted water spreads diseases such as cholera, typhoid and gastro, kills fish and aquatic plants, and makes water unfit for drinking, washing and farming.

(x) How the untreated water can be made fit for drinking?

Untreated water can be made fit for drinking by first filtering it through cloth or sand to remove suspended matter, then boiling it or treating it with chlorine or UV light to kill germs. If it is hard, it can be softened, and if it is muddy it can be settled with alum so impurities sink. The treated water should be stored in clean, covered containers to keep it safe.

Chapter 16: Chemical Industries

Exercise (Chapter 16)

A. Fill in the blanks. (Filling word shown in red.)

(i) Metallurgy deals with **extraction** of pure metals from its ores.

(ii) Solvay process is used for the preparation of **soda ash**.

(iii) Wohler prepared **urea** in 1828.

(iv) Petroleum is believed to be prepared by the action of bacteria upon **organic matter**.

(v) Petroleum is found in the earth in more **depth** than the water.

(vi) Quality of petroleum fuel is determined using a number called **octane number**.

(vii) Chemical formula of Soda ash is **Na2CO3**.

(viii) Ammonium carbamate is the hydrated form of **urea**.

(ix) The first step in metallurgy of a metal is the **crushing** of metal ore.

(x) Petra is a Greek word meaning **rock**.

B. Choose the correct answer. (Correct option marked with a green tick.)

(i) The soda ash is prepared commercially by using:

(a) NaNO3 and KCO3

(b) NaNO3 and NH3

(c) NaCl and MgCO3

(d) NaCl and CaCO3 ✓

(ii) The first step in metallurgical operations is to:

(a) Concentrate the ore

(b) Crushing the ore ✓

(c) Extraction of metal from ore

(d) Refining the ore

(iii) Commercially urea is prepared by:

(a) Carbamide

(b) Ammonia and carbon dioxide ✓

(c) Cyanogens

(d) All of these

(iv) Petroleum is refined by the technique of:

(a) Distillation ✓

(b) Crystallization

(c) Condensation

(d) All of above

(v) As the number of carbon atoms increase in petroleum fractions, it also increases the:

(a) Boiling point

(b) Density

(c) Both of these ✓

(d) None of these

(vi) Solvay process is used to prepare:

(a) Urea

(b) Soda Ash ✓

(c) Gasoline

(d) Iron

(vii) Methane gas is used in preparation of:

(a) Urea

(b) Ammonia gas ✓

(c) Petrol

(d) Copper ore

(viii) LPG contains the mixture of:

(a) Methane and pentane

(b) Methane and ethane

(c) Ethane and Pentane

(d) Propane and butane ✓

(ix) The Major fraction in the CNG is:

(a) Methane ✓

(b) Ethane

(c) Propane

(d) Butane

(x) Octane number is used to measure:

(a) Number of 'C' atoms in a molecule of a fuel

(b) Number of 'H' atoms in the fuel molecules

(c) Efficiency of fuel ✓

(d) Percentage of Methane in any fuel

C. Reasoning and descriptive questions.

(i) Why the Petroleum fuels are called Fossil fuels?

Petroleum fuels are called fossil fuels because they formed from the fossil remains of plants and animals buried under the earth millions of years ago. Under heat, pressure and bacterial action, the buried organic matter slowly changed into coal, petroleum and natural gas, so these fuels are the stored remains of ancient life.

(ii) Diesel provides more energy per mole than petrol.

Diesel provides more energy per mole than petrol because diesel molecules are larger hydrocarbons with more carbon and hydrogen atoms per molecule than petrol molecules.

Burning one mole of diesel therefore breaks and forms more bonds and releases more heat energy, which is why diesel engines give more mileage per litre than petrol engines.

(iii) Extraction of metals from their ore is different for different metals.

Extraction differs because metals differ in reactivity and in the kind of ore they occur in. Reactive metals such as sodium and aluminium are extracted by electrolysis of their molten compounds, moderately reactive metals such as iron and zinc by reduction with carbon in a furnace, and unreactive metals such as gold and silver are found nearly free and need only concentration and refining.

(iv) Why Soda ash is essential to prepare on large scale?

Soda ash is essential on a large scale because it is a basic raw material for many industries: it is used to make glass, soaps and detergents, paper and textiles, and chemicals such as sodium bicarbonate and washing soda. It is also used to soften hard water and in the manufacture of dyes and medicines, so almost every country needs it in bulk.

(v) Urea is more used in Pakistan rather than ammonia fertilizer.

Urea is more used in Pakistan than ammonia fertilizer because urea is a solid, easy to store, transport and spread on fields, while ammonia is a gas that needs pressure tanks and special equipment. Urea also contains 46% nitrogen, the highest of common fertilizers, it is cheaper to make from Pakistan natural gas, and it dissolves in soil moisture and releases nitrogen slowly to crops.

(vi) How the industries help in the progress of mankind?

Industries help the progress of mankind by turning raw materials into useful goods: food, clothes, houses, medicines, machines and transport. They provide jobs and income, raise the standard of living, supply fertilizers that increase food production, and produce the steel, cement, chemicals and energy on which modern life depends. They also fund research, education and better health care.

(vii) Write a brief note on the metallurgical operations? How the copper can be extracted from the earth?

Metallurgical operations are the steps used to obtain a pure metal from its ore: crushing and grinding the ore, concentrating it to remove gangue, extracting the metal by reduction, roasting or electrolysis, and finally refining it to remove impurities. Copper is extracted from its sulphide ore, copper pyrites: the ore is crushed, concentrated by froth flotation, roasted to copper oxide, reduced with carbon or by self-reduction to blister copper, and refined electrolytically to pure copper.

(viii) Discuss the importance of soda ash. Also explain the solvay process for its preparation.

Soda ash, sodium carbonate Na2CO3, is important for making glass, soaps and detergents, paper, textiles and many chemicals, and for softening water. In the Solvay process, brine (NaCl solution) is saturated with ammonia, then carbon dioxide is passed through it to give

sodium bicarbonate, which precipitates: $NaCl + NH_3 + CO_2 + H_2O \rightarrow NaHCO_3 + NH_4Cl$. The bicarbonate is heated to give soda ash: $2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$. The ammonia is recovered from ammonium chloride with slaked lime and reused, and the raw materials are common salt and limestone.

(ix) Why the fertilizers are needed for more production of crops? Explain the preparation method for the urea.

Fertilizers are needed because crops remove nitrogen, phosphorus and potassium from the soil, and without replacing them the soil becomes poor and yields fall. Fertilizers supply these nutrients in soluble form and increase production. Urea is prepared commercially from ammonia and carbon dioxide under high pressure and temperature: first ammonium carbamate forms, $2NH_3 + CO_2 \rightarrow NH_2COONH_4$, and then it loses water to give urea, $NH_2COONH_4 \rightarrow NH_2CONH_2 + H_2O$.

(x) How the petroleum originated in the earth crust? What are the different fractions obtained from the petroleum, and also describe the utilization of these fractions.

Petroleum originated from the remains of tiny marine plants and animals buried under mud millions of years ago; under heat, pressure and bacterial action, the organic matter changed into liquid petroleum and natural gas trapped in porous rocks. By fractional distillation it gives: refinery gases (fuel, LPG), gasoline or petrol (cars), kerosene (cooking, jets), diesel (trucks, tractors), lubricating oil (machinery), wax (candles, polish) and bitumen (roads, roofing). Natural gas, mainly methane, is used for cooking, heating and making fertilizers.